How Random is Human–AI Interaction?

A Next-Token Entropy Study of Human-Imposed Prompt Constraint

1 Literature Review

LLM output as a temperature-driven phase transition. A recent line of work treats the text generated by an LLM as a physical system undergoing an order–disorder transition. Arnold, Holtorf, Schäfer and Lörch [1] adapt statistical-physics methods for automated phase-transition detection to LLMs, computing distributional distances directly on the next-token probability distribution and detecting transitions as a function of temperature, prompt, and training checkpoint. This is the starting point for our study: the next-token distribution shows phase-transition behavior that physics-style order parameters can pick up. In as-yet-non-peer-reviewed work, Nakaishi, Nishikawa and Hukushima [2] report numerically that GPT-2 passes from a low-temperature repetitive regime to a high-temperature incomprehensible one through a genuine critical point near the default temperature $T \approx 1$ (a model- and method-specific value rather than a universal constant), and Sun and Haghighat [3] recast the Transformer as an O(N) spin model with transitions tied to generation temperature. Together these make temperature the standard control parameter and motivate the order/disorder language we use, while leaving open what plays that role when temperature is held fixed.

The softmax as a Gibbs/Boltzmann measure. The connection to statistical mechanics is exact. Ackley, Hinton and Sejnowski [4] introduced the Boltzmann machine, whose equilibrium distribution over states is the Gibbs/Boltzmann form $\pi(x) \propto e^{-E(x)/T}$ with an explicit computational temperature — the same Gibbs/Boltzmann form $\pi(x) = \frac{1}{Z} e^{-E(x)}$. Grathwohl et al. [5] make the modern statement: any softmax classifier is secretly an energy-based model in which the logits define an energy and the softmax is the corresponding Gibbs distribution over tokens. Guo et al. [6] complete the link to our order parameter by showing that temperature scaling, dividing logits by T before the softmax, controls the sharpness and hence the Shannon entropy of that distribution. This is what lets us make the central modeling choice of the paper: the LLM's next-token output is a Boltzmann measure, and we treat its entropy as a physical order parameter.

Shannon entropy and per-token surprisal as the measurement. Our order parameter is Shannon entropy [7], $H = -\sum_i p_i \log_2 p_i$, the canonical measure of the uncertainty of a probability distribution. Farquhar, Kossen, Kuhn and Gal [8] show that this quantity tracks LLM behavior and changes with the question asked: their semantic entropy, Shannon entropy computed over clusters of meaning-equivalent generations, detects hallucinations because narrow factual questions give low entropy while open or unanswerable questions give high entropy. We measure raw next-token entropy, not their semantic entropy; the two can diverge because a model can be sure of the next token yet uncertain about what its answer means, so we read their result as motivation, not as an equivalence. This narrow-versus-open pattern is what our content-constraint axis varies. Meister et al. [9] support the per-token view, modeling generation as a discrete stochastic process in which human-like text keeps per-token surprisal close to the model's conditional entropy, while Holtzman et al. [10] link the low-entropy regime to repetitive, degenerate ("ordered") text, the behavioral signature of the ordered phase from Thread 1.

The human prompt as a control parameter. Sclar, Choi, Tsvetkov and Suhr [11] show that meaning-preserving changes in prompt formatting alone produce large, measurable swings in model behavior, variation that is not eliminated by more examples, larger models, or instruction tuning, which justifies treating the format of the prompt as a control parameter that shifts the output distribution. Lu et al. [12] reinforce this, showing that the mere ordering of in-context examples can move performance from state-of-the-art to chance, and notably use entropy statistics to select prompts, tying prompt

manipulation directly to output entropy. This is why we vary human-imposed constraint, both the content of the question and the format of the system prompt, as the two axes of the experiment.

Concentration of measure. Finally, reasoning quantitatively about a large random system needs concentration-of-measure tools. Chatterjee [13] develops concentration and superconcentration and their links to Markov semigroups, which gives the rigorous basis for why an empirical mean of a random quantity concentrates around its expectation. Bartlett, Foster and Telgarsky [14] apply exactly this machinery (bounded-differences/McDiarmid-type concentration) to neural networks, justifying its use on systems of this kind. We use this directly in Section 4, where Markov's and Chebyshev's inequalities bound the error of our Monte-Carlo entropy estimates.

2 Specific Research Goals

The literature review establishes two results that have so far been pursued separately. First, the next-token distribution of a language model undergoes an order–disorder transition as a function of its sampling temperature, detectable through statistical-physics order parameters computed directly on that distribution [1]. Second, meaning-preserving changes to the prompt measurably shift the output distribution [11], and the Shannon entropy of that distribution is a meaningful, question-dependent quantity [8]. What this literature does not do is place the human in the control-parameter role: temperature is a property of the model's sampling, not of the interaction, so studies that vary temperature answer "how random is the model" rather than the question, "how random is human–AI interaction."

Motivated by this gap, we hold temperature fixed and treat the human's prompt itself as the control parameter, using the Shannon entropy of the next-token distribution as the order parameter. We pursue three specific goals:

1. **Validate content constraint as a control parameter.** Determine whether tightening what is asked, moving a question from open-ended to narrowly factual, monotonically lowers next-token entropy, establishing the human's question as a usable knob on the order parameter.

2. **Characterize the format axis and its interaction with content.** Determine whether and how constraining how the model may answer (e.g. "one word only" versus "respond freely") shapes entropy, and in particular whether the two axes interact, yielding a two-dimensional phase diagram $H(\text{content}, \text{format})$ rather than two independent effects. This two-axis phase diagram is the study's main novel contribution.

3. **Re-center the assigned question on the human.** By varying human constraint at fixed temperature, give a quantitative, information-theoretic answer that locates the randomness of the interaction in the human's prompt rather than in the model's sampling.

Because the entropy is that of a Gibbs measure (Section 3), these goals also connect to the link between optimization and probability: a format constraint acts as an additional constraint on the maximum-entropy program underlying the softmax, so asking how constraint reshapes H is asking how the human reshapes that optimization landscape.

3 Probabilistic Model and Methods

The next-token distribution as a Gibbs measure. At each step a language model produces a vector of logits $z = (z_1, \ldots, z_V)$ over its vocabulary of size V, and samples the next token from the temperature-scaled softmax

$$p_T(i \mid h) \;=\; \frac{\exp(z_i/T)}{\sum_{j=1}^{V} \exp(z_j/T)}, \tag{1}$$

where h is the conversation history so far and $T > 0$ is the sampling temperature. Writing $E_i = -z_i$ for the energy of token i and $\beta = 1/T$ for the inverse temperature, (1) is exactly the Gibbs / Boltzmann measure

$$p_T(i \mid h) \;=\; \frac{1}{Z}\, e^{-\beta E_i}, \qquad Z = \sum_{j=1}^{V} e^{-\beta E_j},$$

the same form $\pi(x) = \frac{1}{Z} e^{-E(x)}$, with the logits playing the role of (negative) energies and the softmax denominator the partition function Z. This is what lets us use statistical-mechanics language: the model's output at a given step is a probability measure over an energy landscape, and temperature controls how sharply that mass concentrates on the lowest-energy (highest-logit) tokens [6]. We hold $T = 1$ fixed throughout, so that temperature is not a variable in this study; the control parameters we vary are imposed by the human, as described below.

Entropy as the order parameter. For a fixed history h we summarize the next-token measure (1) by its Shannon entropy [7],

$$H(h) \;=\; -\sum_{i=1}^{V} p_T(i \mid h)\, \log_2 p_T(i \mid h) \qquad \text{(bits)}. \tag{2}$$

H is our order parameter: it is near 0 when the model is effectively certain of the next token (mass concentrated on one token, the "ordered" regime) and large when many tokens are comparably likely (mass spread out, the "disordered" regime). In practice we compute p_T from the API's top-K log-probabilities[1] ($K = 20$, renormalized to sum to one), so measured entropy lies in $[0, \log_2 20]$ bits; the top-K truncation underestimates H uniformly across conditions and so does not bias the comparisons below.

Two human-imposed control parameters. Rather than vary temperature, we ask how two axes of human constraint shape H, which re-centers the question on the human side of the interaction [11]:

- **Content constraint** of the question, varied within each conversation. Each turn of a fixed script is labelled narrow (a factual question with one intended answer), medium, or open (an open-ended question with many acceptable answers). Narrow questions are expected to concentrate the next-token measure and open questions to spread it [8].
- **Format constraint** of the system prompt, varied between conditions. Three system prompts impose increasingly loose output formats: tight ("answer in one word only"), medium, and loose ("respond freely"). This axis constrains how the model may answer rather than what is being asked.

The two axes give a 3×3 design with a two-dimensional phase diagram of H as a function of (content, format).

Multi-turn measurement protocol. We use a single fixed ten-turn human script (turns $t = 0, \ldots, 9$) whose content labels are held identical across all conditions, preceded by a baseline turn ($t = -1$) consisting of the system prompt and a narrow first question with no prior history. For each turn we query the API for the first ten response tokens together with their top-20 log-probabilities, compute the entropy (2) of each token's distribution, and average these to obtain a single per-turn value H_t. We average over the first ten tokens, rather than the first token alone, so that the measurement reflects the uncertainty of the answer as a whole and not merely the model's choice of an opening word; the one trade-off this introduces is examined as a limitation in Section 4. We then sample a

[1] Code availability: `https://github.com/John-P-Tobin/Math_60_Final_Project`.

separate full response and append it to the history, so that conversational state accumulates from turn to turn. The dialogue is a stochastic process in which each H_t is measured on the realized history up to turn t.

Replication and estimation. We run the full ten-turn script 20 times independently for each of the three format conditions, giving $20 \times 3 \times 10 = 600$ per-turn measurements. Because the per-turn values within a single run share an accumulating history, they are not independent; the independent unit is the *run*. For inference we therefore collapse each run to one mean per (content $\times$ format) cell, yielding $n = 20$ independent observations per cell. We treat these as an i.i.d. Monte-Carlo estimate of the cell-mean entropy $\mathbb{E}[H]$ and report its standard error, a Chebyshev concentration bound on the estimation error (both developed in Section 4), and Welch two-sample t-tests for differences across format conditions.

4 Preliminary Results

All results use the per-token Shannon entropy H (2) as the order parameter, measured against the two human-imposed control parameters of Section 3: the content constraint of the question (narrow / medium / open) and the format constraint of the system prompt (tight / medium / loose). Following that protocol, the unit of statistical replication is the run ($n = 20$ per format condition): because turns within a run share an accumulating history and are not independent, we collapse each run to a single mean per (content $\times$ format) cell, on which all standard errors and tests below are computed.

The content axis behaves as a clean control parameter. Within every format condition, mean entropy increases monotonically across the content axis (Fig. 1), with no exceptions among the nine cells: tight rises $0.00 \to 0.64 \to 1.21$ bits, medium $0.03 \to 0.43 \to 0.83$, and loose $0.19 \to 0.47 \to 0.76$ from narrow to medium to open content. At the level of raw next-token entropy this mirrors the behaviour reported for semantic-level uncertainty by [8]: questions with a single intended answer concentrate probability mass, whereas open-ended questions spread it. The monotonicity validates content constraint as a usable control parameter on H.

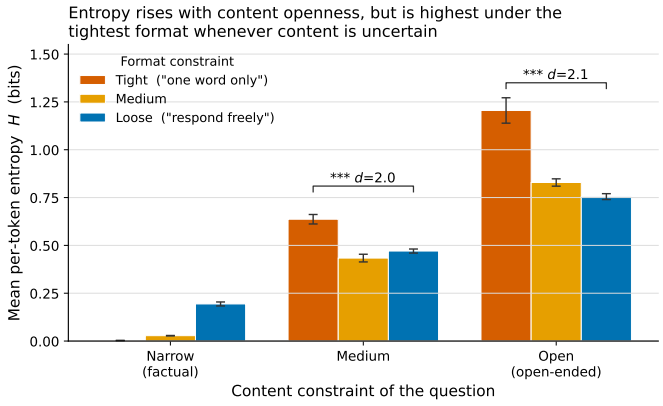


Figure 1: Mean per-token entropy H by content constraint (question) and format constraint (system prompt). Content is monotone in every format condition; the format axis is non-monotone, with the tightest format producing the highest H whenever content is uncertain. Bars are means over 20 independent runs; error bars are ± 1 SE. *** denotes Holm-adjusted $p < 10^{-4}$ (Welch), Cohen's $d \approx 2$.

The format axis is non-monotone and counterintuitive. The central finding is that the tightest format produces the highest per-token entropy whenever the content is uncertain. For open content, tight (1.21 bits) exceeds both medium (0.83) and loose (0.76); for medium content, tight (0.64) exceeds medium (0.43) and loose (0.47). Welch two-sample t-tests on the run-level means confirm all four

contrasts (Table 1): every Holm-adjusted $p < 2 \times 10^{-5}$ with Cohen's $d \approx 2$, and the shift is consistent rather than outlier-driven — 18 of the 20 tight/open runs exceed the single largest loose/open run. Mann–Whitney U tests agree on all four, ruling out a normality artifact.

We read this through the maximum-entropy view of the softmax. A format constraint such as "one word only" does not reduce the model's uncertainty about what to say; it removes the downstream tokens over which that uncertainty would otherwise have been distributed, forcing the entire choice into a single high-entropy token. Constraint therefore redistributes uncertainty rather than removing it. The optimization-side reading is that tightening the format adds constraints to a maximum-entropy program whose probability mass is reallocated, not eliminated [6].

Content	Comparison	Means (bits)	Holm-adj. p	Cohen's d
Open	tight vs. loose	1.21 vs. 0.76	4.5×10^{-6}	2.09
Open	tight vs. medium	1.21 vs. 0.83	1.8×10^{-5}	1.72
Medium	tight vs. loose	0.64 vs. 0.47	4.5×10^{-6}	1.95
Medium	tight vs. medium	0.64 vs. 0.43	8.5×10^{-7}	2.01

Table 1: Format-axis non-monotonicity. Welch two-sample t-tests on run-level means ($n = 20$ per cell), Holm–Bonferroni corrected across the four comparisons. The tightest format produces significantly higher per-token entropy than the looser formats whenever content is uncertain.

On factual content the format effect reverses sign. The redistribution picture is sharpened by the narrow-content column, where the format ordering flips: loose is highest (0.19 bits) and tight collapses to near zero (0.003). For a question with one correct answer, "respond freely" lets the model spend entropy on optional preamble and hedging, whereas "one word only" forces the single determined token. The sign of the format effect thus depends on content, it amplifies entropy when the answer is genuinely open and suppresses it when the answer is fixed, which is exactly what a redistribution of uncertainty predicts.

Turn-by-turn dynamics. Figure 2 tracks H_t across the ten-turn script. Entropy follows the scripted content label turn by turn, spiking on open turns and dropping to near zero on narrow turns. The tight-format excess is largest at the first open question ($t = 1$, $H \approx 1.89$ bits versus ≈ 0.91 for the looser formats) and relaxes over later turns. We interpret this as the accumulating conversation state, itself a Markov process over the next-token distribution, progressively narrowing the set of plausible single-word answers, so that even a tight format has less residual uncertainty to concentrate as the dialogue proceeds [10].

Quantifying estimation error via concentration. The twenty runs constitute an i.i.d. Monte-Carlo estimate $\bar{H}_n$ of the true cell mean $\mathbb{E}[H]$, so Chebyshev's inequality bounds the estimation error in the same form as the Monte-Carlo integration bound:

$$P\big(\big|\bar{H}_n - \mathbb{E}[H]\big| \geq \varepsilon\big) \;\leq\; \frac{\mathrm{Var}(H)}{n\,\varepsilon^2}.$$

Inverting for a 95% guarantee ($\delta = 0.05$, $n = 20$) gives a distribution-free half-width of exactly one sample standard deviation. Even this conservative bound separates the conditions: for tight/open it yields [0.91, 1.50] bits, entirely above the loose/open interval [0.69, 0.82]. The corresponding Central Limit Theorem half-widths are roughly 2.3× tighter; we report the Chebyshev interval as the rigorous, distribution-free guarantee, with the CLT version as its asymptotic refinement [13].

Limitation. Because H is averaged over token positions, the tight-format estimate could in principle be inflated if that condition emits fewer content-bearing tokens, leaving fewer low-entropy tokens to

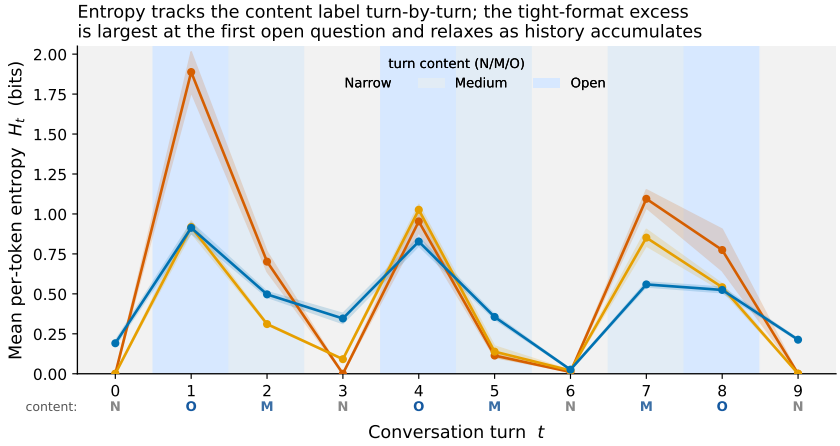


Figure 2: Per-turn entropy H_t across the fixed ten-turn script (background shaded by each turn's scripted content label). Entropy tracks content turn-by-turn; the tight-format excess is largest at the first open question ($t = 1$) and relaxes as conversation history accumulates.

dilute the average. Two observations argue against a pure averaging artifact: the excess is largest precisely at $t = 1$, and the competing failure mode, early stopping into low-entropy padding, would push H down, not up. The right check is a position-matched comparison of first-token entropy across formats, which we set out as the immediate next step in Section 5.

5 Future Directions

Our preliminary results suggest four future steps, each building directly on a finding above.

Settle the averaging limitation with a position-matched probe. The most immediate step is the first-token comparison flagged in Section 4: re-running a small probe on the open-content turns that records the entropy of each individual token position rather than only the per-turn average, together with the number of content-bearing tokens generated. If the tight format still shows higher entropy at token position 0 alone, the redistribution effect is a genuine distributional fact; if the gap appears only in the average, it is an artifact of token count. This is cheap to run and would convert the headline result from strongly suggestive to confirmed.

Add temperature as a third axis. We deliberately held $T = 1$ to isolate human constraint, but the phase-transition literature [1] varies temperature instead. A natural extension is the full three-axis surface $H(\text{content}, \text{format}, T)$, testing whether tighter human prompts shift the critical temperature at which the output transitions from ordered to disordered. This would connect our human-centered control parameters directly to the existing temperature-driven order–disorder picture.

Explain the sign reversal on factual content. The narrow-content reversal deserves its own treatment. A targeted experiment varying only factual questions across formats, while measuring where in the response the entropy is spent (opening token, hedging, qualifiers), would test the proposed mechanism that constraint redistributes rather than removes uncertainty.

Model the turn-by-turn relaxation as a Markov process. The decay of the tight-format excess as history accumulates (Section 4) invites an explicit stochastic-process treatment: the conversation state evolves as a Markov chain over the next-token distribution, and the relaxation of H_t toward a content-determined floor resembles convergence toward a stationary regime. Estimating that relaxation rate, and whether it depends on the format constraint, would let us model the dynamics explicitly as a Markov chain converging to a stationary regime.

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